

TASK 1 – 2D VIRTUAL TRY-ON RESEARCH & PROTOTYPE DEVELOPMENT

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Date: 25/05/2026

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Objective

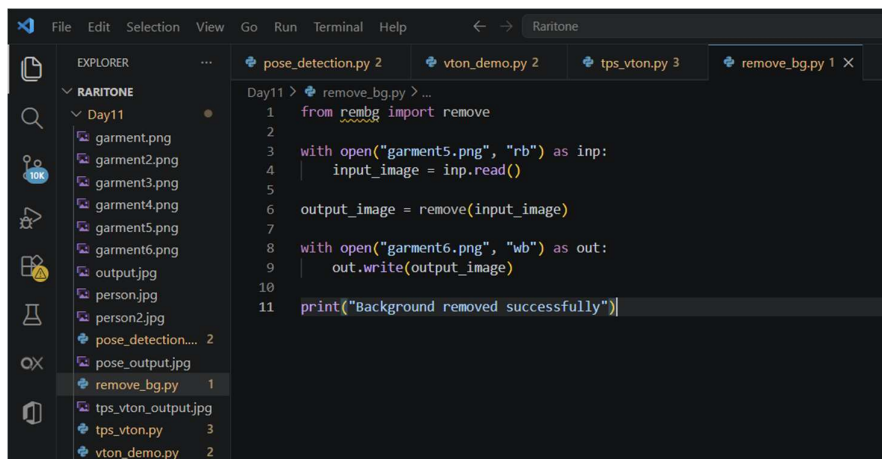
To develop a basic 2D Virtual Try-On prototype capable of digitally placing a garment image onto a person's image using pose detection and image processing techniques.

Introduction

Virtual Try-On (VTON) is a computer vision technology that enables users to visualize clothing digitally without physically wearing the garment. It is widely used in e-commerce and fashion technology applications to improve customer experience and assist in clothing selection.

In this task, a basic 2D Virtual Try-On prototype was developed using OpenCV and MediaPipe. The prototype detects body landmarks from an uploaded image and places a garment image on the detected torso region to simulate a virtual clothing trial.

Folder structure:



Technologies Used

Python

Python was used as the primary programming language for implementing image processing and pose estimation functionalities.

OpenCV

OpenCV was used for:

- Reading images
- Resizing images
- Image manipulation
- Clothing overlay
- Output generation

MediaPipe

MediaPipe Pose was used for:

- Human pose estimation
- Body landmark detection
- Skeleton visualization
- Shoulder coordinate extraction

NumPy

NumPy was used for:

- Matrix operations
- Coordinate transformations
- TPS transformation calculations
- Image array processing

rembg

rembg was used for:

- Background removal
- Transparent garment generation
- Garment preprocessing

Input Images

Person Image

A full-body standing image was used as input for pose detection and clothing placement.

Characteristics:

- Single person
- Front-facing pose
- Full body visible
- Clear background



Garment Image

A shirt image was used as the clothing item for virtual try-on simulation.

Characteristics:

- Front view shirt image
- Suitable dimensions for overlay
- Used for torso alignment



Garment Preprocessing

Before performing Virtual Try-On, the garment image was preprocessed to improve overlay quality.

Operations performed:

- Background removal
- Transparent PNG generation
- Garment cleaning
- Image normalization

Tool Used:

rembg

Output:



Benefits:

- Removes unwanted background pixels
- Produces cleaner overlay results
- Improves garment fitting appearance
- Enables transparency-based blending

Background Removal Implementation

Code:

```
from rembg import remove
```

```
with open("garment2.png","rb") as inp:  
    image = inp.read()
```

```
output = remove(image)
```

```
with open("garment4.png","wb") as out:  
    out.write(output)
```

Generated Output:



Workflow

Person Image



Image Loading



Image Preprocessing



Pose Detection

↓
Shoulder Landmark Extraction
↓
Garment Loading
↓
Background Removal
↓
TPS Style Garment Warping
↓
Transparent Overlay
↓
Virtual Try-On Output

Step 1 – Image Loading

The person image and garment image were loaded using OpenCV.

Code:

```
person = cv2.imread("person.jpg")  
cloth = cv2.imread("garment.png")
```

Purpose:

- Load input images
- Prepare images for processing

Step 2 – Image Resizing

Images were resized to maintain consistent dimensions and improve display quality.

Code:

```
person = cv2.resize(  
    person,  
    (700,1000)  
)
```

Purpose:

- Standardize image size
- Improve visualization
- Simplify garment placement

Step 3 – Pose Detection

MediaPipe Pose was used to detect body landmarks.

Code:

```
results = pose.process(rgb)
```

Detected landmarks include:

- Nose
- Eyes
- Shoulders
- Elbows
- Wrists
- Hips
- Knees
- Ankles

Total landmarks detected:

33 body landmarks

Step 4 – Skeleton Visualization

Detected landmarks were visualized using MediaPipe drawing utilities.

Code:

```
mp_draw.draw_landmarks(  
    image,  
    results.pose_landmarks,  
    mp_pose.POSE_CONNECTIONS  
)
```

Output:

- Skeleton structure
- Joint connections
- Body landmark points



Step 5 – Shoulder Detection

The left and right shoulder landmarks were extracted.

Code:

```
left_shoulder = lm[11]  
right_shoulder = lm[12]
```

Purpose:

- Determine torso position
- Estimate clothing placement region
- Align garment image

Step 6 – Garment Placement

The garment image was resized and positioned on the torso region using shoulder coordinates.

Code:

```
person[  
    y_offset:y_offset + 320,  
    x_offset:x_offset + 260  
] = cloth
```

Purpose:

- Simulate clothing fitting
- Generate virtual try-on effect
- Visualize garment placement

Step 7 – TPS Style Garment Warping

To improve garment alignment, a TPS-style transformation was applied before overlaying the garment onto the person image.

Purpose:

- Simulate garment deformation
- Improve body alignment
- Adjust garment shape
- Create a more natural Virtual Try-On appearance

Code:

```
src = np.float32([
    [0,0],
    [w,0],
    [0,h],
    [w,h]
])
```

```
dst = np.float32([
    [15,10],
    [w-15,0],
    [0,h],
    [w,h-15]
])
```

```
matrix = cv2.getPerspectiveTransform(
    src,
    dst
)
```

```
warped = cv2.warpPerspective(
    cloth,
    matrix,
    (w,h)
)
```

Output:



Step 8– Output Generation(Demo)

The final image was generated and saved.

Code:

```
cv2.imwrite(
    "vton_output.jpg",
    person
)
```

Output:

- Virtual Try-On image
- Garment overlay result
- Skeleton visualization



Results

- Background removal
- Transparent garment generation
- TPS style garment transformation
- Garment warping
- Improved Virtual Try-On visualization

Observations

- Pose detection accuracy directly affects garment placement.
- Higher image quality improves landmark detection.
- Proper garment sizing improves visualization.
- Front-facing images produce better results.
- MediaPipe provides reliable landmark extraction for basic Virtual Try-On systems.
- Modern Virtual Try-On systems combine pose estimation, segmentation models, and garment warping techniques to improve realism.

Future Improvements

The current prototype performs direct garment overlay. Future enhancements may include:

- Human body segmentation
- AI-based garment fitting
- Deep learning Virtual Try-On models (CP-VTON, VITON-HD, HR-VITON)
- Body mesh reconstruction and 3D cloth simulation for advanced Virtual Try-On systems
- Evaluation using fashion datasets such as DeepFashion2 and VITON-HDReal-time
- Virtual Try-On systems

Conclusion

A basic 2D Virtual Try-On prototype was successfully developed using OpenCV and MediaPipe. The system detects human body landmarks, identifies the torso region, and overlays a garment image to simulate digital clothing trials. This prototype serves as a foundational step toward developing advanced AI-powered Virtual Try-On systems with realistic garment fitting and body-aware transformations.